mance and to protect the system components from damage PCM will disengage the A/C compressor clutch when high side pressure rises above 2971 kPa (431 psi) or fall below 206 kPa (30 psi). The A/C pressure transducer input to the PCM will also prevent the A/C compressor clutch from engaging when ambient temperatures are below about 1.0° C (50° F) due to the pressure/temperature relationship of the refrigerant.

The A/C pressure transducer is diagnosed using a DRBIII® scan tool. Refer to the appropriate diagnostic information.

The A/C pressure transducer cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

A/C PRESSURE TRANSDUCER

The A/C pressure transducer is tested using a DRBIII® scan tool. Refer to the appropriate diagnostic information. Before testing the A/C pressure transducer, be certain that the transducer wire harness connection is clean of corrosion and properly connected. For the A/C to operate, an A/C pressure transducer voltage reading between 0.451 and 4.519 volts is required. Voltages outside this range indicate a low or high refrigerant system pressure condition to the powertrain control module (PCM). The PCM is programmed to respond to a low or high refrigerant system pressure by suppressing operation of the A/C compressor. Refer to the A/C Pressure Transducer Voltage chart for the possible conditions indicated by the transducer voltage reading.

A/C PRESSURE TRANSDUCER VOLTAGE

Voltage	Possible Indication
0.0	1. No sensor supply voltage from PCM. 2. Shorted sensor circuit. 3. Faulty transducer.
0.150 TO 0.450	1. Ambient temperature below 10° C (50° F). 2. Low refrigerant system pressure.
0.451 TO 4.519	1. Normal refrigerant system pressure.
4.520 TO 4.850	1. High refrigerant system pressure.
5.0	1. Open sensor circuit. 2. Faulty transducer.

REMOVAL

NOTE: Note: It is not necessary to discharge the refrigerant system to replace the A/C pressure transducer.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C pressure transducer (Fig. 21).

(3) Remove the A/C pressure transducer from the A/C discharge line and remove and discard the O-ring seal.

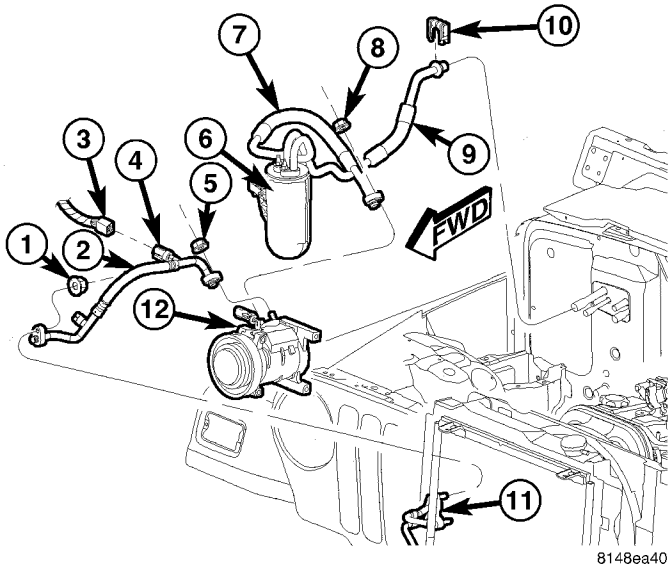

Fig. 21 A/C Pressure Transducer - 2.4L Engine

1 - NUT
2 - A/C DISCHARGE LINE
3 - WIRE HARNESS CONNECTOR
4 - A/C PRESSURE TRANSDUCER
5 - NUT
6 - ACCUMULATOR
7 - A/C SUCTION LINE (FRONT SECTION)
8 - NUT (4)
9 - A/C SUCTION LINE (REAR SECTION)
10 - SECONDARY RETAINING CLIP
11 - A/C CONDENSER
12 - A/C COMPRESSOR

INSTALLATION

NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C pressure transducer fitting.

(2) Install the A/C pressure transducer onto the A/C discharge line. Hand-tighten the transducer securely.